香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

General Chemistry CHM1001

Electrochemistry (Ch. 20)

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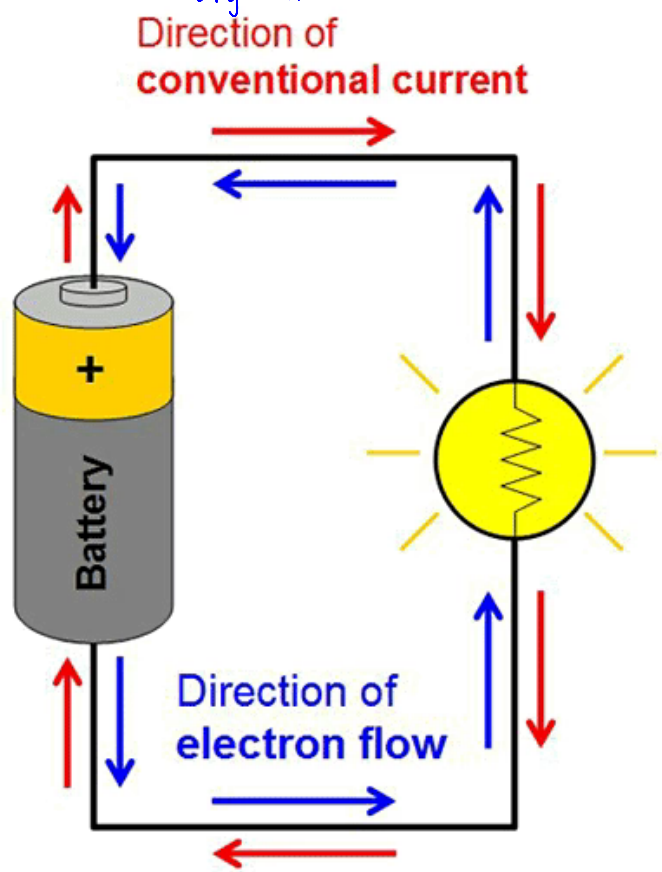
Electrochemistry

Electrochemistry is the study of the relationships between electricity and chemical reactions. It includes the study of both spontaneous and nonspontaneous processes.

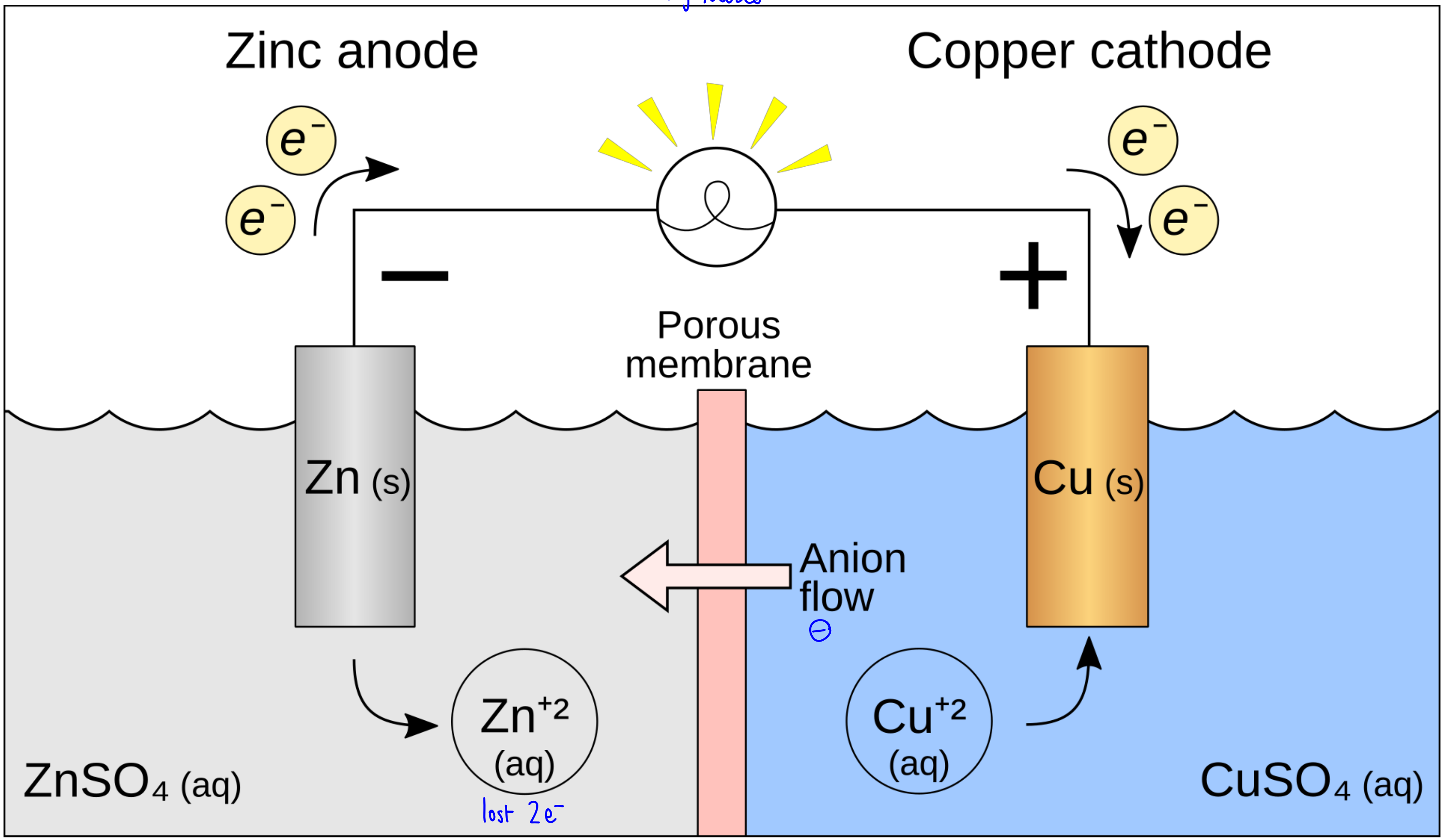
Battery

Dry cell



Battery

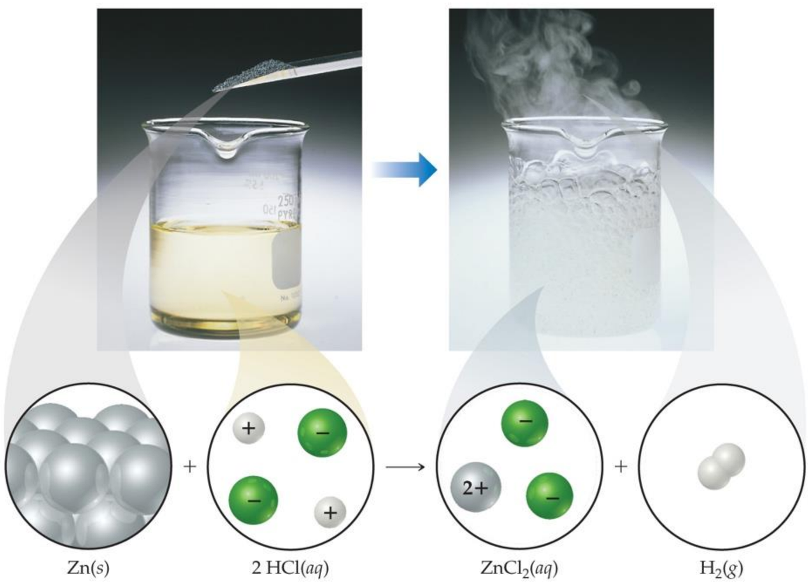
Mainly focused



Synopsis of Assigning Oxidation Numbers (as a Reminder)

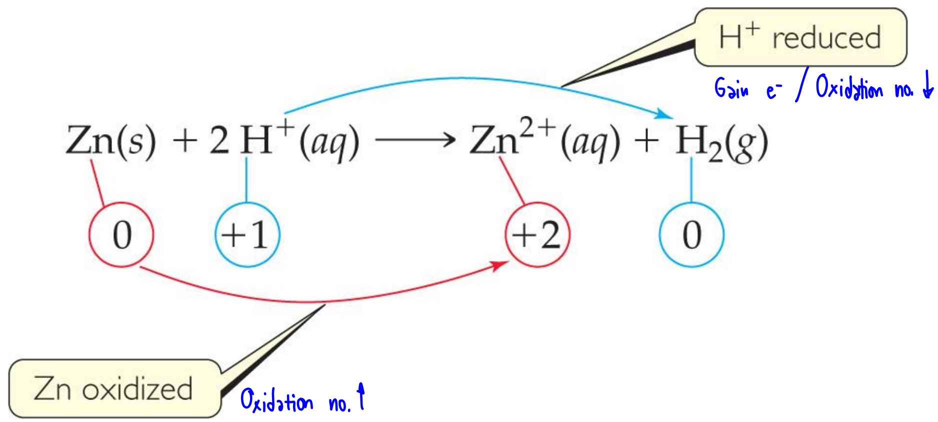
1. Elements = 0
2. Monatomic ion = charge
3. F: –1
4. O: –2 (unless peroxide = –1)
5. H: +1 (unless a metal hydride = –1)
6. The sum of the oxidation numbers equals the overall charge (0 in a compound).

Oxidation Numbers



- To keep track of what loses electrons and what gains them, we assign **oxidation numbers**.
- If the oxidation number increases for an element, that element is oxidized.
- If the oxidation number decreases for an element, that element is reduced.

Oxidation and Reduction



- A species is **oxidized** when it loses electrons.
  - Zinc loses two electrons, forming the Zn^{2+} ion.
- A species is **reduced** when it gains electrons.
  - H^+ gains an electron, forming H_2.
- An **oxidizing agent** causes something else to be oxidized (H^+); a **reducing agent** causes something else to be reduced (Zn).

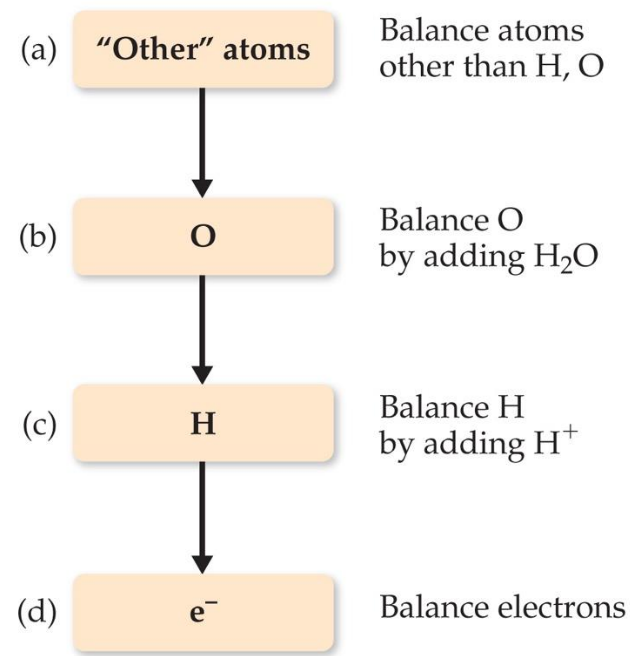
Half-Reactions

- The oxidation and reduction are written and balanced separately.
- We will use them to balance a redox reaction.
- For example, when Sn^{2+} and Fe^{3+} react,

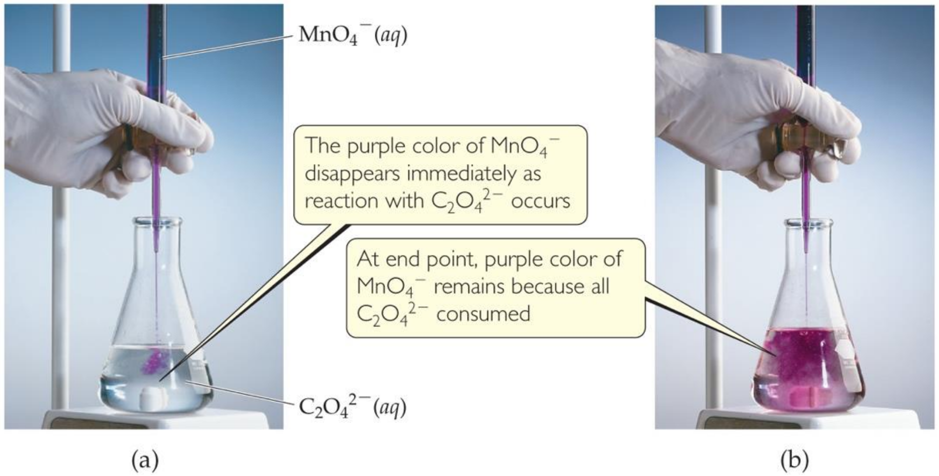
$$\textit{Oxidation:} \quad Sn^{2+}(aq) \longrightarrow Sn^{4+}(aq) + 2e^-$$
$$\textit{Reduction:} \quad 2\,Fe^{3+}(aq) + 2e^- \longrightarrow 2\,Fe^{2+}(aq)$$

Balancing Redox Equations: The Half-Reactions Method (a Synopsis)

1) Make two half-reactions (oxidation and reduction).
2) Balance atoms other than O and H. Then, balance O and H using H_2O/H^+.
3) Add electrons to balance charges.
4) Multiply by common factor to make electrons in half-reactions equal.
5) Add the half-reactions.
6) Simplify by dividing by common factor or converting H^+ to OH^- if basic.
7) Double-check atoms and charges balance!



The Half-Reaction Method



Consider the reaction between MnO_4^- and $C_2O_4^{2-}$:

7+ 2- ×4 = 8- +3×2=+6 -2×4=-8 +4

$$MnO_4^-(aq) + C_2O_4^{2-}(aq) \longrightarrow Mn^{2+}(aq) + CO_2(aq)$$

- Assigning oxidation numbers shows that Mn is reduced ($+7 \rightarrow +2$) and C is oxidized ($+3 \rightarrow +4$).

Oxidation Half-Reaction

$$C_2O_4^{2-} \longrightarrow CO_2$$

To balance the carbon, we add a coefficient of 2:

$$C_2O_4^{2-} \longrightarrow 2\ CO_2$$

Oxidation Half-Reaction

$$C_2O_4^{2-} \longrightarrow 2\ CO_2$$

The oxygen is now balanced as well. To balance the charge, we must add two electrons to the right side:

$$C_2O_4^{2-} \longrightarrow 2\ CO_2 + 2e^-$$

Reduction Half-Reaction

$$MnO_4^- \longrightarrow Mn^{2+}$$

The manganese is balanced; to balance the oxygen, we must add four waters to the right side:

$$MnO_4^- \longrightarrow Mn^{2+} + 4\ H_2O$$

Reduction Half-Reaction

$$MnO_4^- \longrightarrow Mn^{2+} + 4\ H_2O$$

To balance the hydrogen, we add $8H^+$ to the left side:

$$8\ H^+ + MnO_4^- \longrightarrow Mn^{2+} + 4\ H_2O$$

Reduction Half-Reaction

$$8\ H^+ + MnO_4^- \longrightarrow Mn^{2+} + 4\ H_2O$$

To balance the charge, we add $5e^-$ to the left side:

$$5e^- + 8\ H^+ + MnO_4^- \longrightarrow Mn^{2+} + 4\ H_2O$$

Combining the Half-Reactions

Now we combine the two half-reactions together:

$$C_2O_4^{2-} \longrightarrow 2\ CO_2 + 2e^- \quad \times 5$$
$$5e^- + 8\ H^+ + MnO_4^- \longrightarrow Mn^{2+} + 4\ H_2O \quad \times 2$$

To make the number of electrons equal on each side, we will multiply the first reaction by 5 and the second by 2:

Combining the Half-Reactions

$$5\ C_2O_4^{2-} \longrightarrow 10\ CO_2 + 10e^-$$

$$10e^- + 16\ H^+ + 2\ MnO_4^- \longrightarrow 2\ Mn^{2+} + 8\ H_2O$$

When we add these together, we get

$$10e^- + 16\ H^+ + 2\ MnO_4^- + 5\ C_2O_4^{2-} \longrightarrow$$
$$2\ Mn^{2+} + 8\ H_2O + 10\ CO_2 + 10e^-$$

Combining the Half-Reactions

$$10e^{-} + 16\ H^{+} + 2\ MnO_4^{-} + 5\ C_2O_4^{2-} \longrightarrow 2\ Mn^{2+} + 8\ H_2O + 10\ CO_2 + 10e^{-}$$

The only thing that appears on both sides is the electrons. Subtracting them, we are left with

$$16\ H^{+} + 2\ MnO_4^{-} + 5\ C_2O_4^{2-} \longrightarrow 2\ Mn^{2+} + 8\ H_2O + 10\ CO_2$$

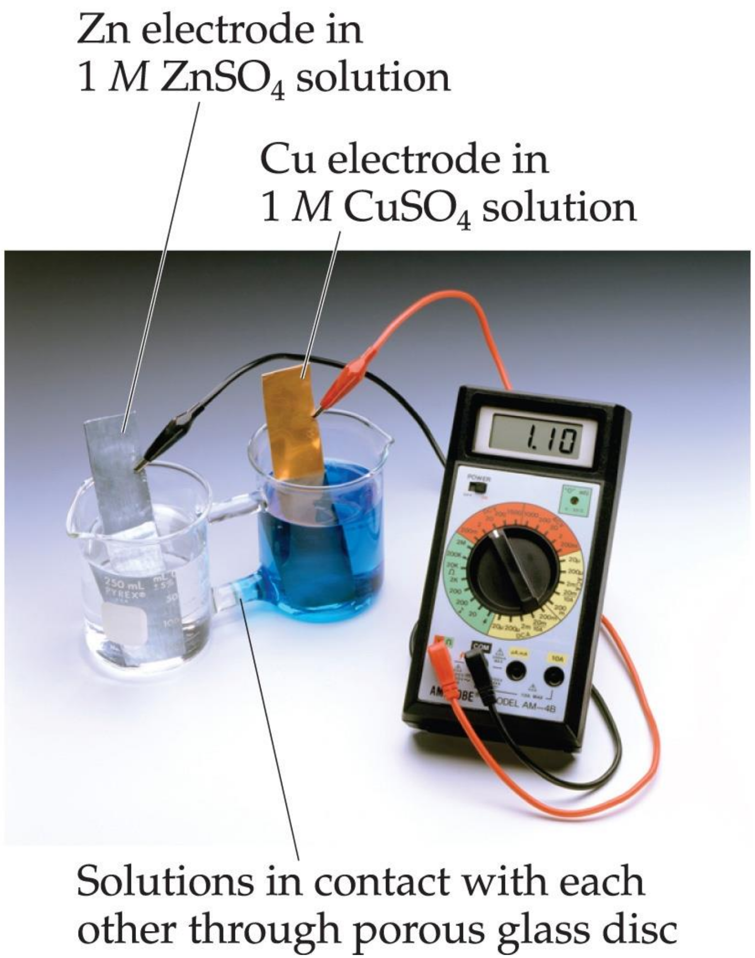
(Verify that the equation is balanced by counting atoms and charges on each side of the equation.)

Balancing in Basic Solution

- A reaction that occurs in basic solution can be balanced as if it occurred in acid.
- Once the equation is balanced, add OH^- to each side to "neutralize" the H^+ in the equation and create water in its place.
- If this produces water on both sides, subtract water from each side so it appears on only one side of the equation.

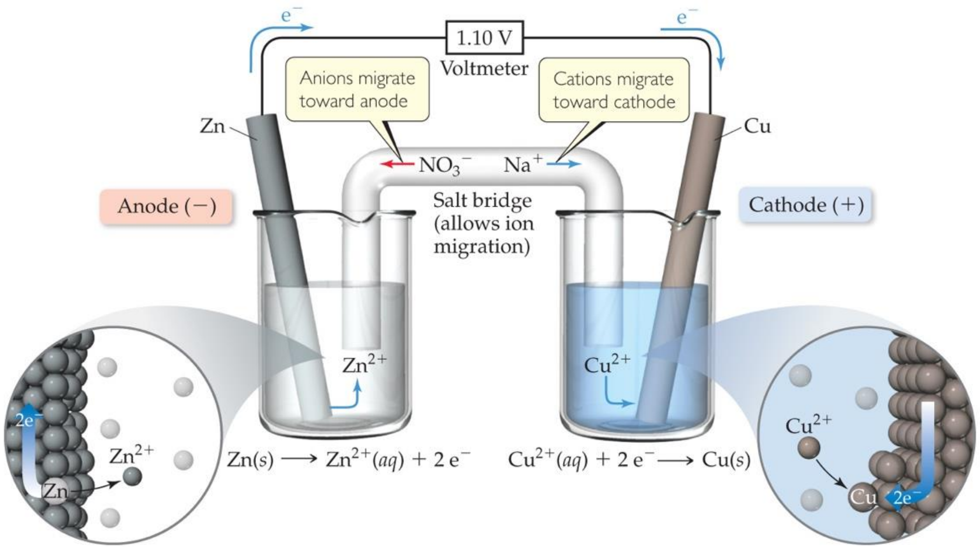
Voltaic Cells (Galvanic cells)

- In spontaneous redox reactions, electrons are transferred and energy is released.
- That energy can do work if the electrons flow through an external device.
- This is a **voltaic cell**.



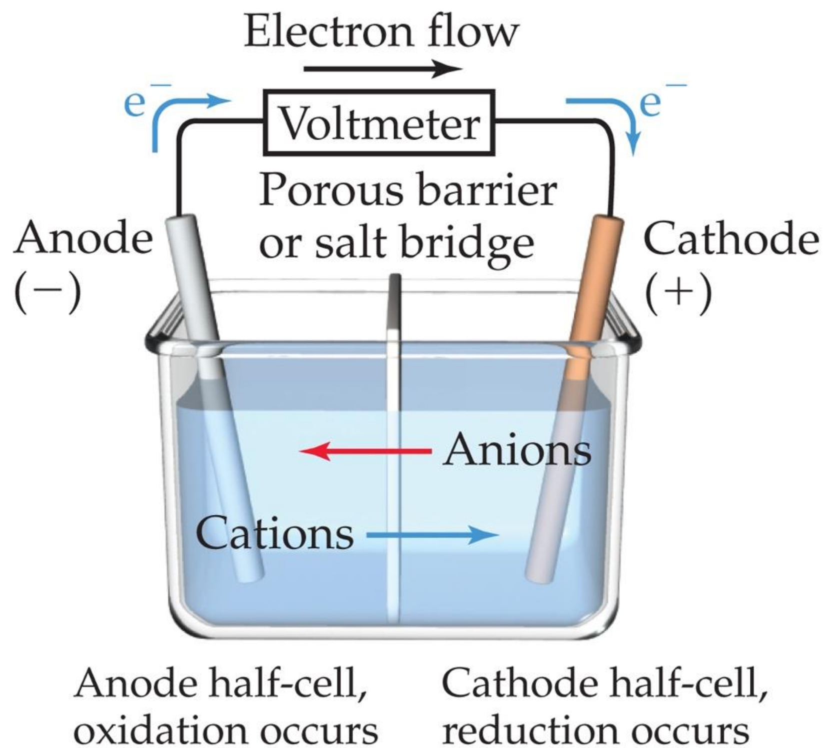
Voltaic Cells (Galvanic cells)

- The oxidation occurs at the **anode**. ⊖ ออก
- The reduction occurs at the **cathode**. ⊖ เข้า
- When electrons flow, charges aren't balanced. So, a salt bridge, usually a U-shaped tube that contains a salt/agar solution, is used to keep the charges balanced.



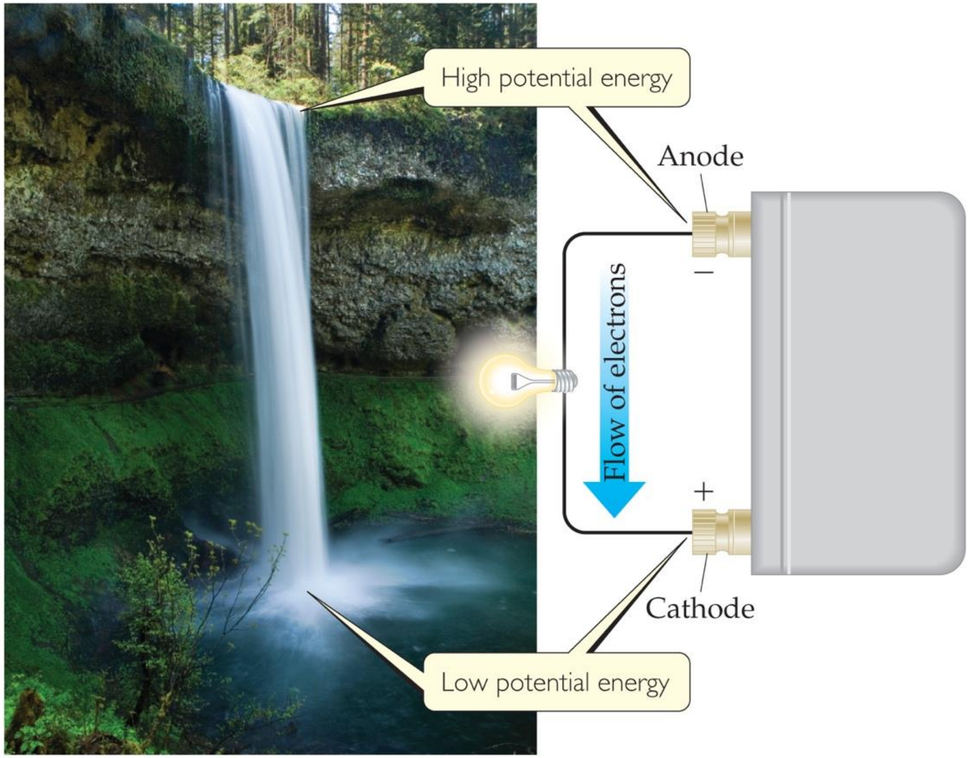
Voltaic Cells (Galvanic cells)

- In the cell, electrons leave the anode and flow through the wire to the cathode.
- Cations are formed in the anode compartment.
- As the electrons reach the cathode, cations in solution are attracted to the now negative cathode.
- The cations gain electrons and are deposited as metal on the cathode.



Electromotive Force (emf)

- Water flows spontaneously one way in a waterfall.
- Comparably, electrons flow spontaneously one way in a redox reaction, from high to low potential energy.



Electromotive Force (emf)

- The potential difference between the anode and cathode in a cell is called the **electromotive force (emf)**.
- It is also called the **cell potential** and is designated E_{cell}.
- It is measured in volts (V). One volt is one joule per coulomb (1 V = 1 J/C).

- Voltage (Volt) is a quantity that represents the energy given to (or lost by) a unit charge.

Standard Reduction Potentials

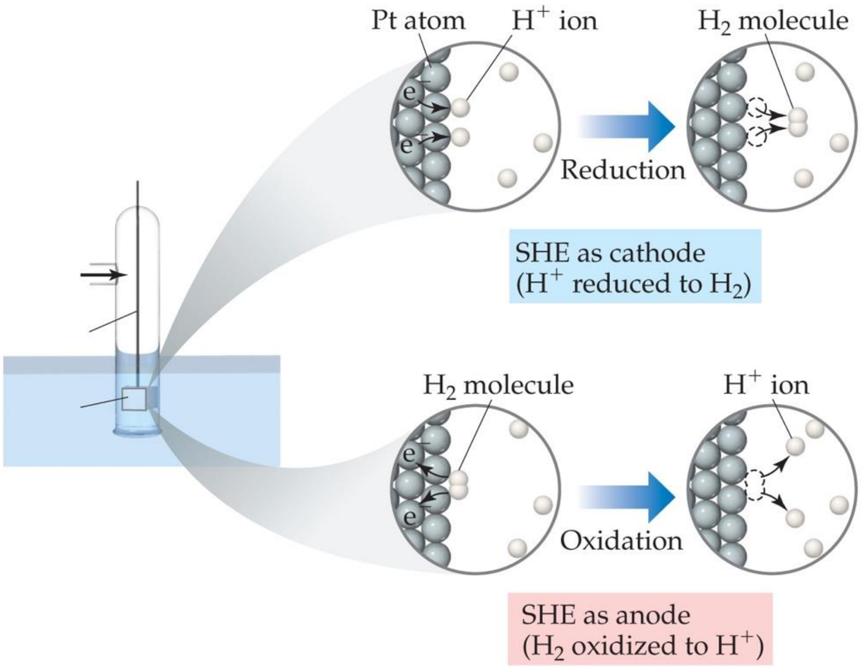
- Reduction potentials for many electrodes have been measured and tabulated.
- The values are compared to the reduction of hydrogen as a standard.

Table 20.1 Standard Reduction Potentials in Water at 25 °C

E°_{red} (V)	Reduction Half-Reaction
+2.87	$F_2(g) + 2e^- \longrightarrow 2F^-(aq)$
+1.51	$MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$
+1.36	$Cl_2(g) + 2e^- \longrightarrow 2Cl^-(aq)$
+1.33	$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$
+1.23	$O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l)$
+1.06	$Br_2(l) + 2e^- \longrightarrow 2Br^-(aq)$
+0.96	$NO_3^-(aq) + 4H^+(aq) + 3e^- \longrightarrow NO(g) + 2H_2O(l)$
+0.80	$Ag^+(aq) + e^- \longrightarrow Ag(s)$
+0.77	$Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$
+0.68	$O_2(g) + 2H^+(aq) + 2e^- \longrightarrow H_2O_2(aq)$
+0.59	$MnO_4^-(aq) + 2H_2O(l) + 3e^- \longrightarrow MnO_2(s) + 4OH^-(aq)$
+0.54	$I_2(s) + 2e^- \longrightarrow 2I^-(aq)$
+0.40	$O_2(g) + 2H_2O(l) + 4e^- \longrightarrow 4OH^-(aq)$
+0.34	$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$
0 [defined]	$2H^+(aq) + 2e^- \longrightarrow H_2(g)$
−0.28	$Ni^{2+}(aq) + 2e^- \longrightarrow Ni(s)$
−0.44	$Fe^{2+}(aq) + 2e^- \longrightarrow Fe(s)$
−0.76	$Zn^{2+}(aq) + 2e^- \longrightarrow Zn(s)$
−0.83	$2H_2O(l) + 2e^- \longrightarrow H_2(g) + 2OH^-(aq)$
−1.66	$Al^{3+}(aq) + 3e^- \longrightarrow Al(s)$
−2.71	$Na^+(aq) + e^- \longrightarrow Na(s)$
−3.05	$Li^+(aq) + e^- \longrightarrow Li(s)$

Standard Hydrogen Electrode

- Their reference is called the standard hydrogen electrode (SHE).
- By definition as the standard, the reduction potential for hydrogen is 0 V:



$$2\ H^+(aq,\ 1M) + 2e^- \longrightarrow H_2(g,\ 1\ \text{atm})$$

Standard Cell Potentials

The cell potential at standard conditions can be found through this equation:

$$E^\circ_{\text{cell}} = E^\circ_{\text{red}}\,(\text{cathode}) - \underset{\text{oxidation}}{E^\circ_{\text{red}}}\,(\text{anode})$$

Because cell potential is based on the potential energy per unit of charge, it is an intensive property.

Standard Cell Potentials

The cell potential at standard conditions can be found through this equation:

$$E^\circ_{cell} = E^\circ_{red}\,(\text{cathode}) - E^\circ_{red}\,(\text{anode})$$

Because cell potential is based on the potential energy per unit of charge, it is an intensive property.

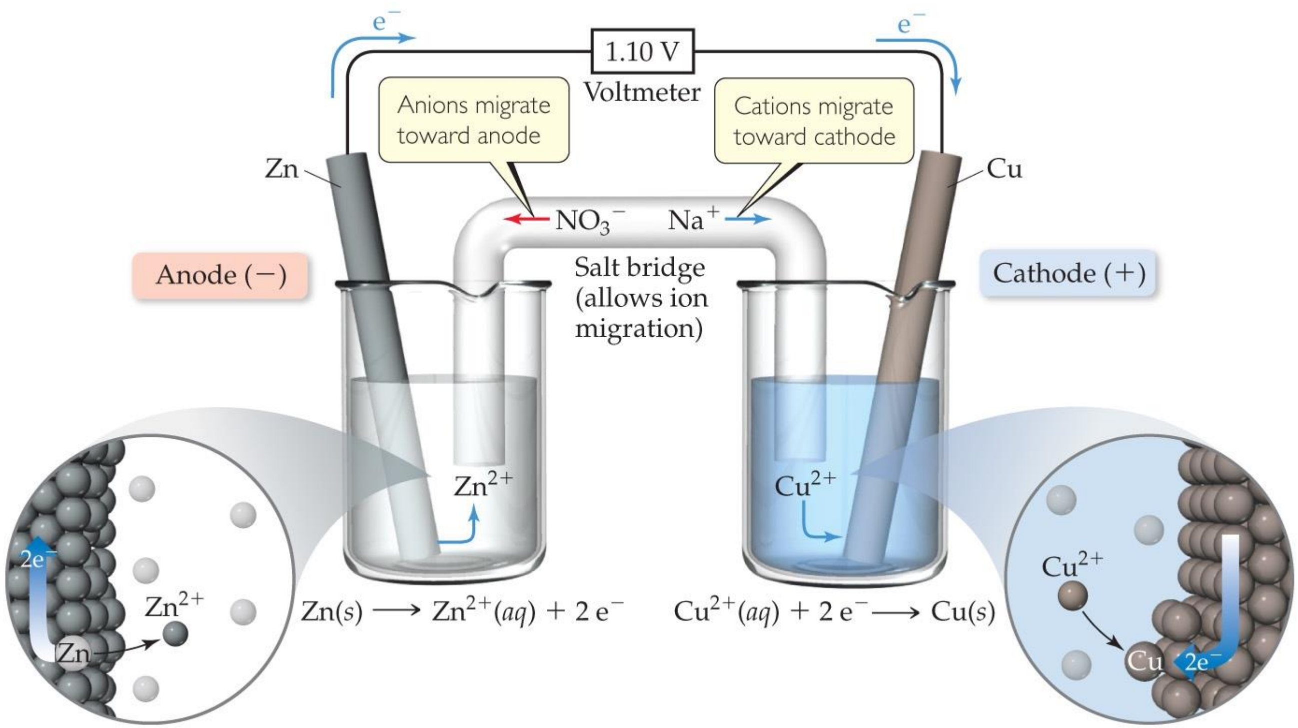
An intensive property is a characteristic that does not depend on the amount of substance, but is determined solely by its type or state.

The cell potential (E° cell) is an intensive property because it represents the energy per unit charge.

This reflects the fact that voltage is a measure of "driving force" or "electrical pressure."

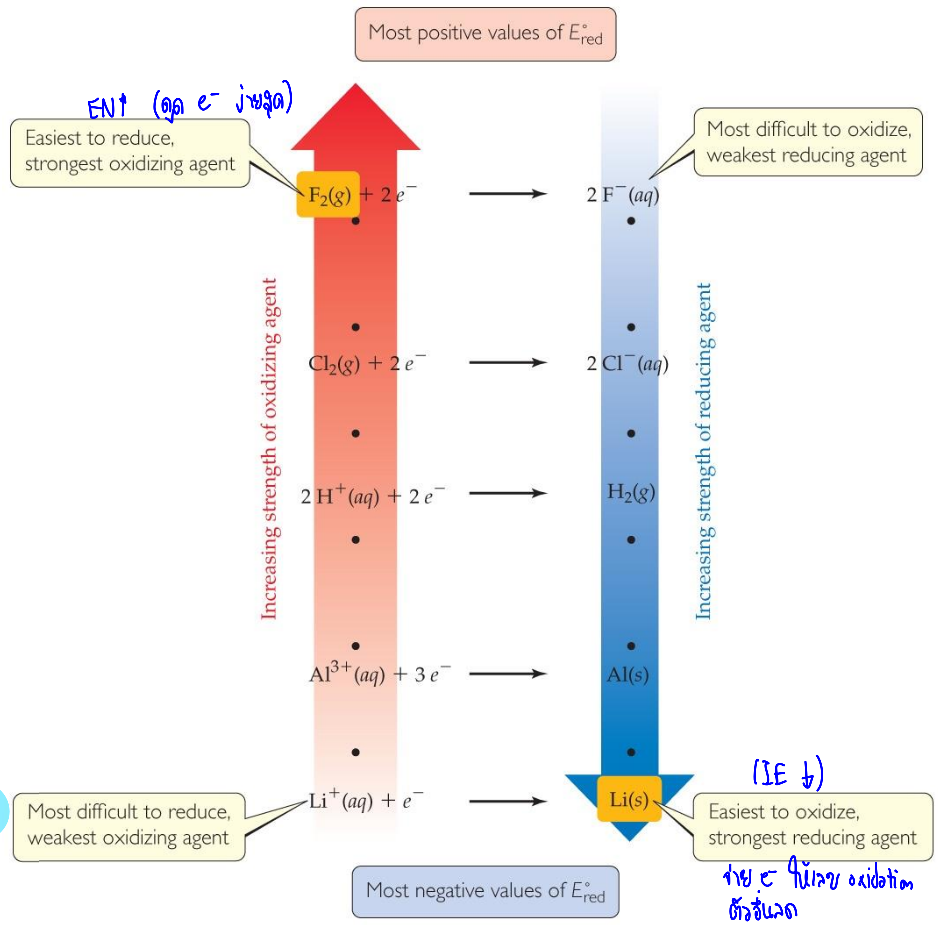
Cell Potentials

- For the anode in this cell, $E°_{red}$ = –0.76 V
- For the cathode, $E°_{red}$ = +0.34 V
- So, for the cell, $E°_{cell} = E°_{red}$ (cathode) – $E°_{red}$ (anode) = +0.34 V – (–0.76 V) = +1.10 V



Oxidizing and Reducing Agents

- The more positive the value of E°_{red}, the greater the tendency for reduction under standard conditions.
- The strongest oxidizers have the most positive reduction potentials.
- The strongest reducers have the most negative reduction potentials.

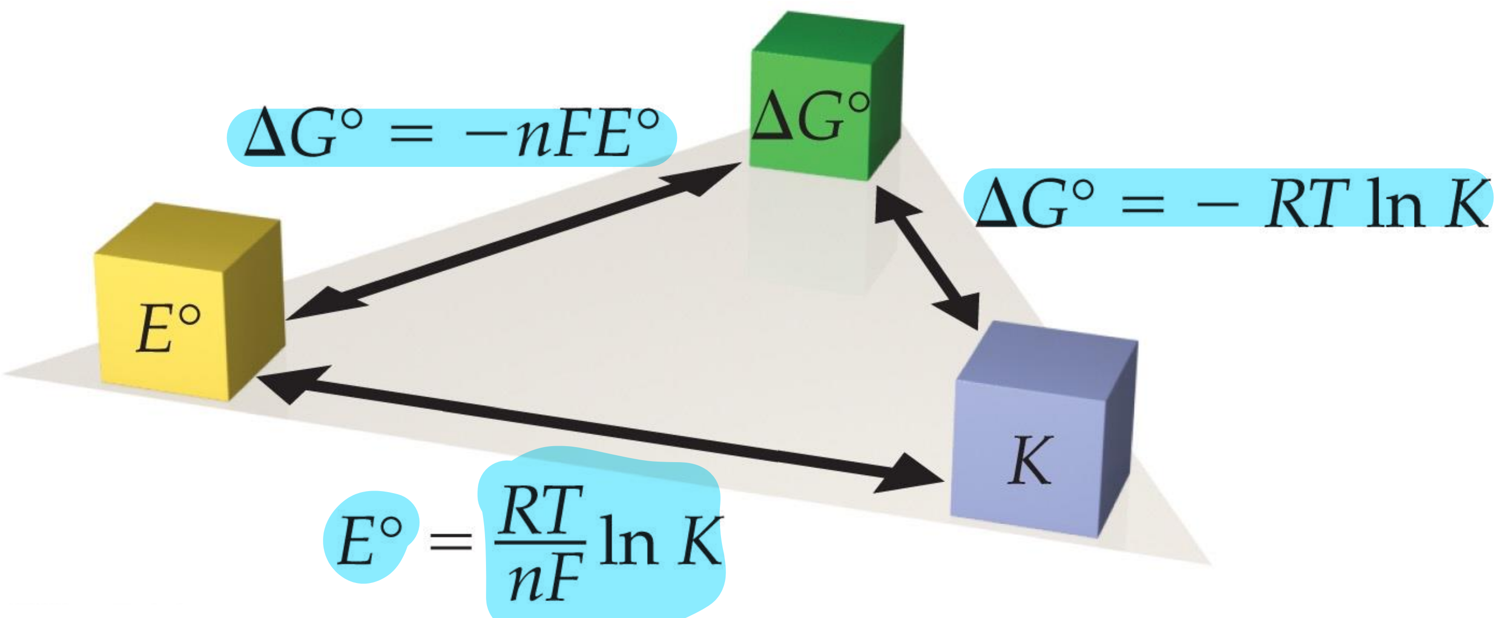


Free Energy and Redox

- Spontaneous redox reactions produce a positive cell potential, or emf.
- $E° = E°_{red}$ (reduction) – $E°_{red}$ (oxidation)
- Note that this is true for ALL redox reactions, not only for voltaic cells.
- Since Gibbs free energy is the measure of spontaneity, positive emf corresponds to negative ΔG.
- How do they relate? $\Delta G = -nFE$ (F is the Faraday constant, 96,485 C/mol.)

Free Energy, Redox, and K

- How is everything related?
- $\Delta G° = -nFE° = -RT \ln K$



Nernst Equation

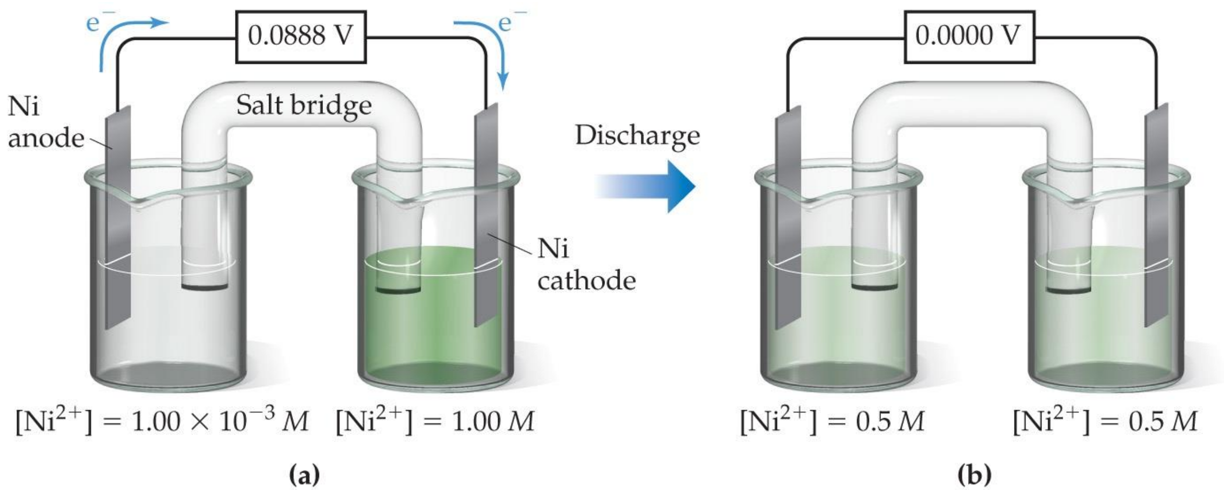
- Remember, $\Delta G = \Delta G^\circ + RT \ln Q$
- So, $-nFE = -nFE^\circ + RT \ln Q$
- Dividing both sides by $-nF$, we get the **Nernst equation**:

$$E = E^\circ - (RT/nF) \ln Q$$

- OR $E = E^\circ - (2.303\ RT/nF) \log Q$
- Using standard thermodynamic temperature and the constants R and F,

$$E = E^\circ - (0.0592/n) \log Q$$

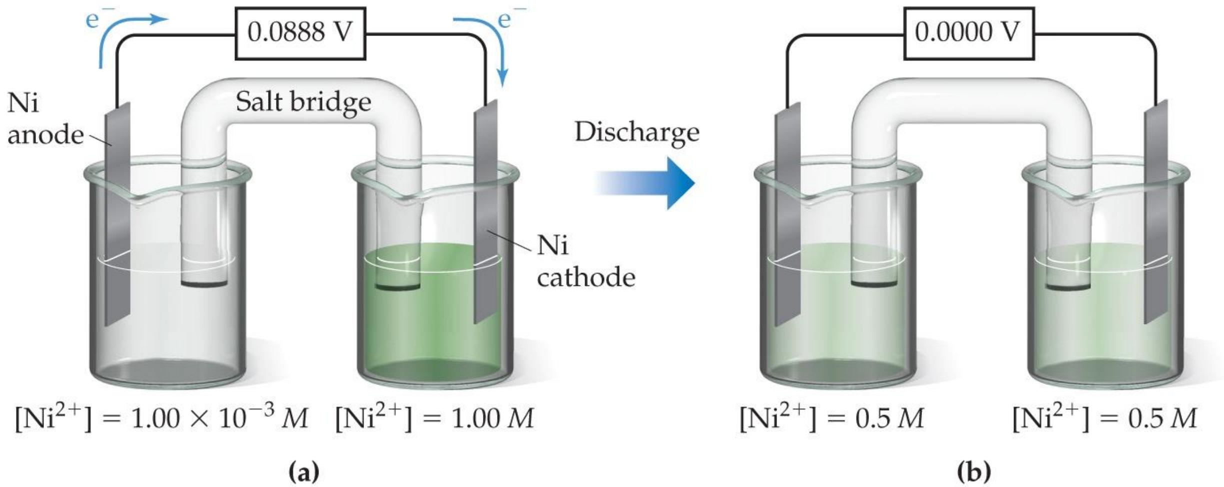
Concentration Cells



Different conc on both sides

- Notice that the Nernst equation implies that a cell could be created that has the same substance at both electrodes, called a **concentration cell**.
- For such a cell, E°_{cell} would be 0, but Q would not.
- Therefore, as long as the concentrations are different, E will not be 0.

Concentration Cells



- **Half-Reactions:**

 Anode (oxidation): $Ni \rightarrow Ni^{2+}(dilute) + 2e^-$ (low-concentration side)

 Cathode (reduction): $Ni^{2+}(concentrated) + 2e^- \rightarrow Ni$ (high-concentration side)

 → **Overall reaction:** $Ni^{2+}(concentrated) \rightarrow Ni^{2+}(dilute)$

- **Nernst Equation:**

 $E = E° - (RT/nF) \ln Q$

Where:

$E°$ = 0 V (because it's a concentration cell)

n = 2 (number of electrons transferred)

$Q = [Ni^{2+}]\_dilute / [Ni^{2+}]\_concentrated = (1.00 \times 10^{-3}) / (1.00) = 1.00 \times 10^{-3}$

T = 298 K (standard temperature)

R = 8.314 J/mol·K

F = 96,485 C/mol

E = E° - (0.0592 / n) log Q

Substituting values:

E = 0 - (0.0592 / 2) × log(1.00 × 10⁻³)

= - (0.0296) × (-3)

*= **+0.0888 V***

Some Applications of Cells

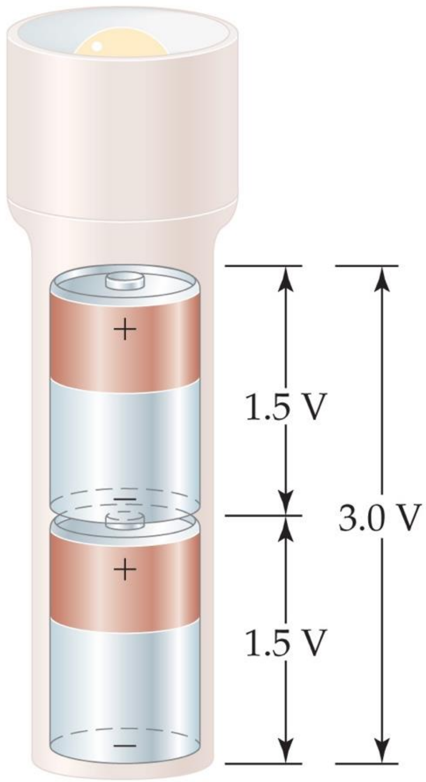
- Electrochemistry can be applied as follows:

❖ Batteries: a portable, self-contained electrochemical power source that consists of one or more voltaic cells.

  - Batteries can be primary cells (cannot be recharged when "dead"—the reaction is complete) or secondary cells (can be recharged).

❖ Prevention of corrosion ("rust-proofing")

❖ Electrolysis



Some Examples of Batteries

- Lead–acid battery: reactants and products are solids, so *Q* is 1 and the potential is independent of concentrations; however, made with lead and sulfuric acid (hazards).
- Alkaline battery: most common primary battery.
- Ni–Cd and Ni–metal hydride batteries: lightweight, rechargeable; Cd is toxic and heavy, so hydrides are replacing it.
- Lithium-ion batteries: rechargeable, light; produce more voltage than Ni-based batteries.

Some Examples of Batteries

- Lead–acid battery: reactants and products are solids, so Q is 1 and the potential is independent of concentrations; however, made with lead and sulfuric acid (hazards).

Electrode Reactions:

- **Anode (oxidation):** $Pb + HSO_4^- \rightarrow PbSO_4 + H^+ + 2e^-$
- **Cathode (reduction):** $PbO_2 + HSO_4^- + 3H^+ + 2e^- \rightarrow PbSO_4 + 2H_2O$

Since **$PbSO_4$ is a solid at both electrodes**, the reaction quotient **Q = 1**, meaning **the cell potential is independent of concentration changes** — according to the Nernst equation, **$E = E°$** under these conditions.

Some Examples of Batteries

- Alkaline battery: most common primary battery.

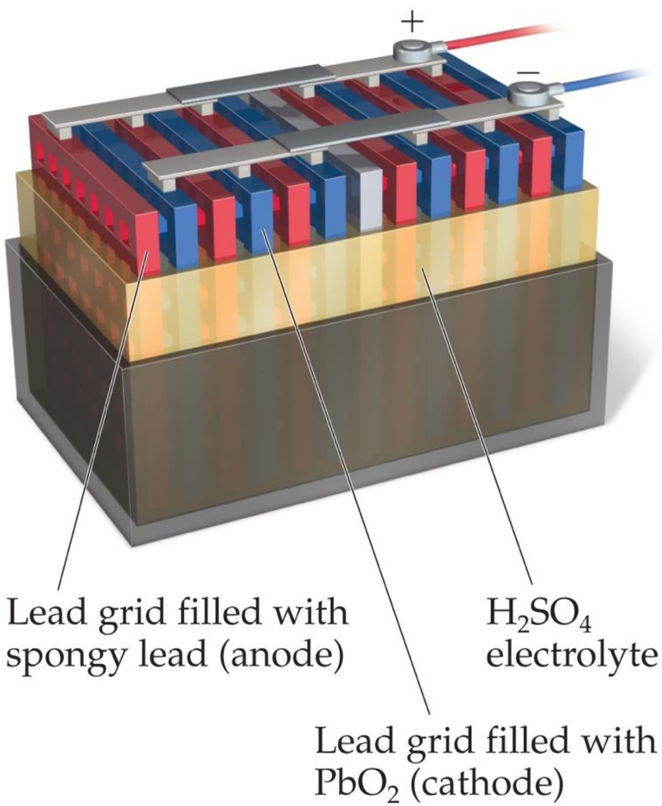
Electrode Reactions:

- **Anode (Zn):** $Zn + 2OH^- \rightarrow ZnO + H_2O + 2e^-$
- **Cathode (MnO_2):** $2MnO_2 + H_2O + 2e^- \rightarrow Mn_2O_3 + 2OH^-$

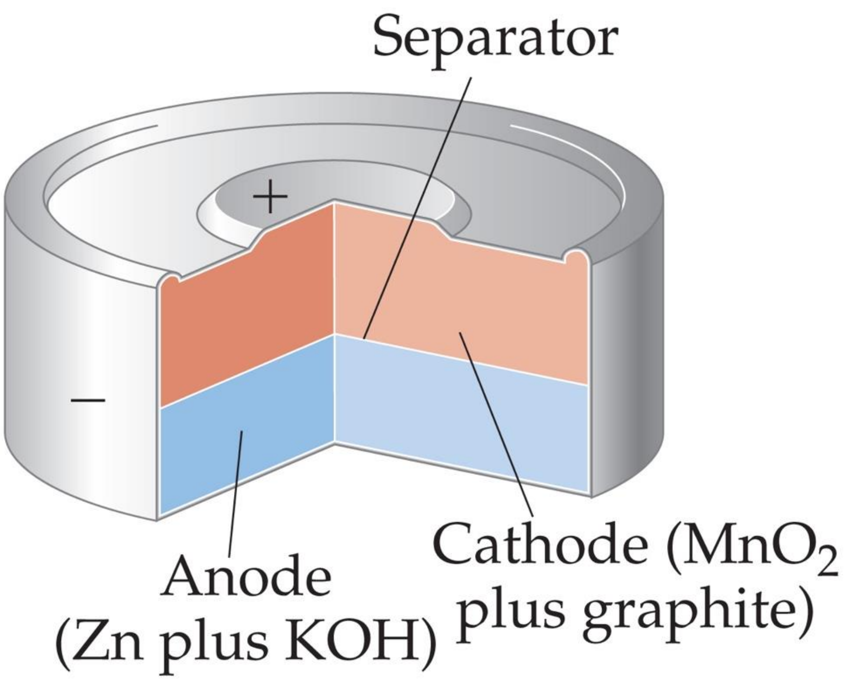
Long-lasting, high capacity, and inexpensive — these are the most common dry batteries in everyday use.
However, **ZnO is poorly soluble in water and adheres to the electrode surface**, so it **cannot revert to metallic Zn during charging**.
Mn_2O_3 is too structurally stable to easily convert back to MnO_2.
Additionally, the **electrolyte (KOH) loses water through evaporation or changes in concentration**, leading to **gas evolution (H_2, O_2) during charging** — which poses a **risk of rupture or leakage**.
→ The reactions are irreversible → Therefore, recharging is not possible

Some Batteries

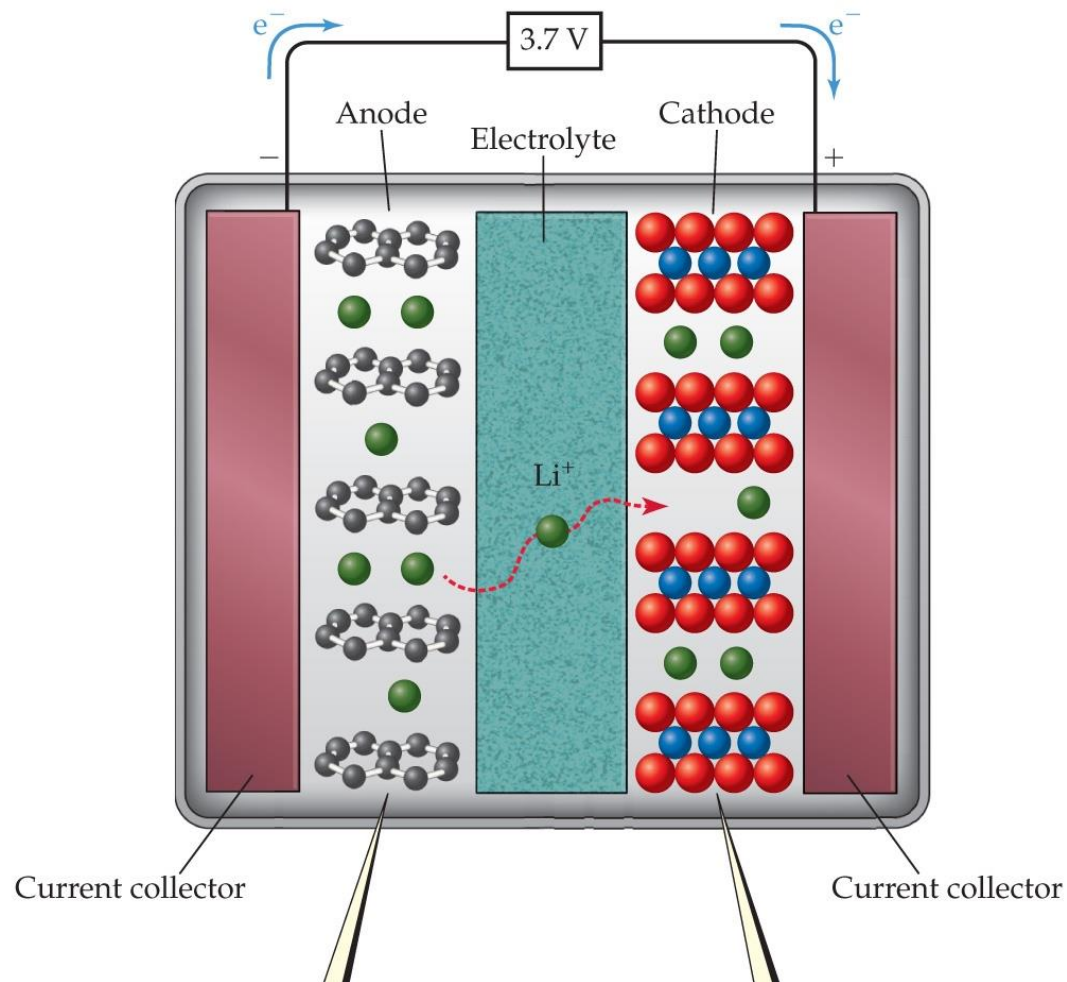
Lead–Acid Battery



Alkaline Battery



Lithium-Ion Battery



The graphite anode contains layers of carbon atoms (black spheres). Li^+ can move in and out of the space between the layers.

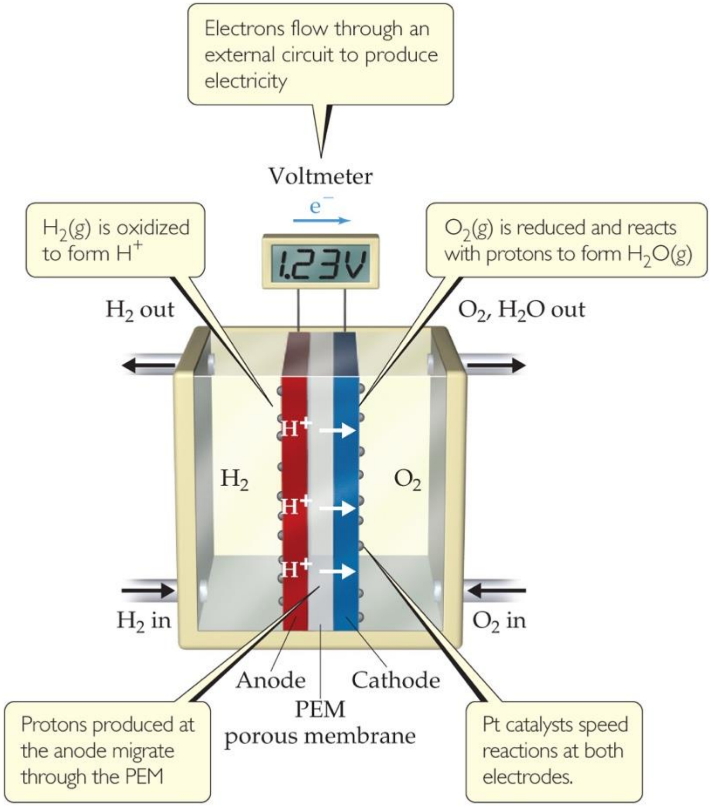
The cathode contains cobalt oxide layers (blue spheres = Co, red spheres = O). Li^+ can move in and out of the space between the layers.

Fuel Cells

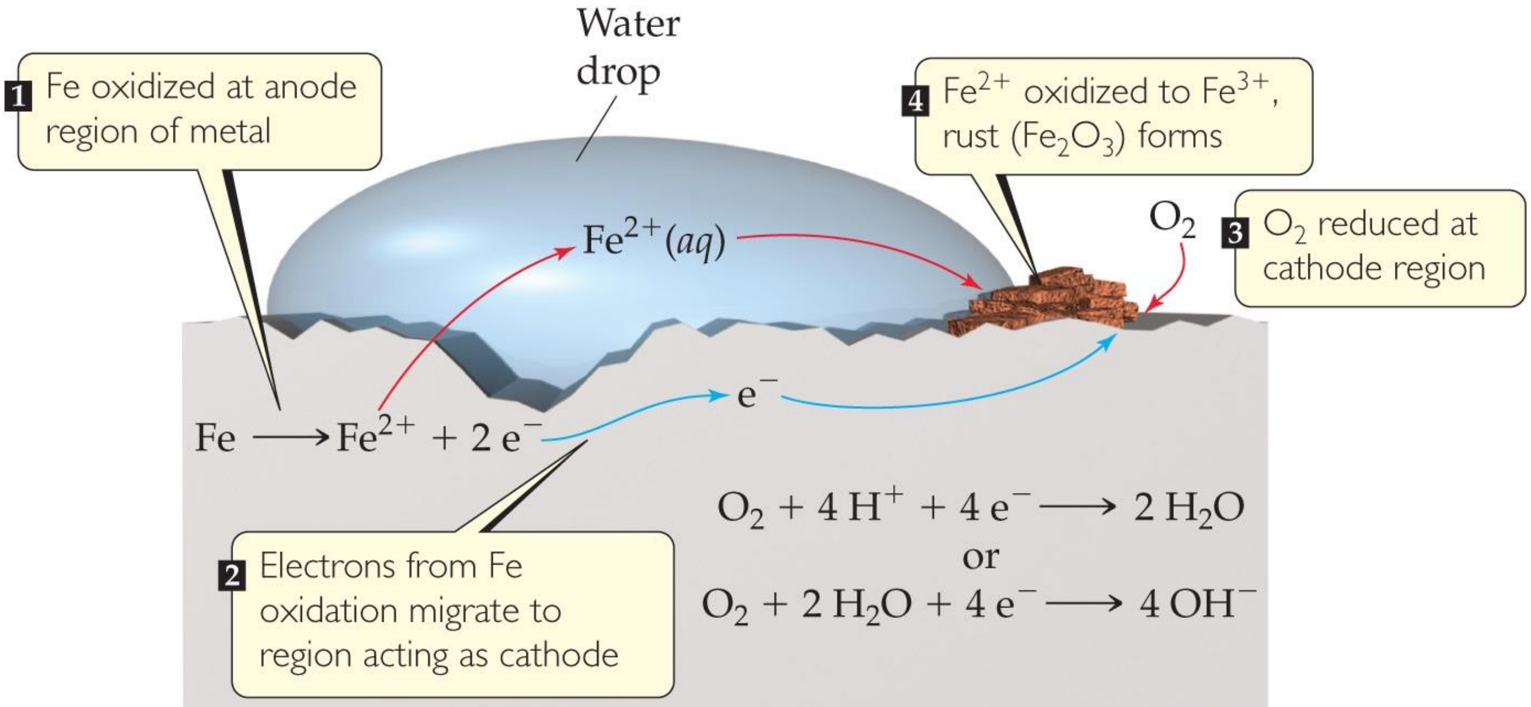
- When a fuel is burned, the energy created can be converted to electrical energy.
- Usually, this conversion is only 40% efficient, with the remainder lost as heat.
- The direct conversion of chemical to electrical energy is expected to be more efficient and is the basis for **fuel cells**.
- Fuel cells are NOT batteries; the source of energy must be continuously provided.

Hydrogen Fuel Cells

- In this cell, hydrogen and oxygen form water.
- The cells are twice as efficient as combustion.
- The cells use hydrogen gas as the fuel and oxygen from the air.

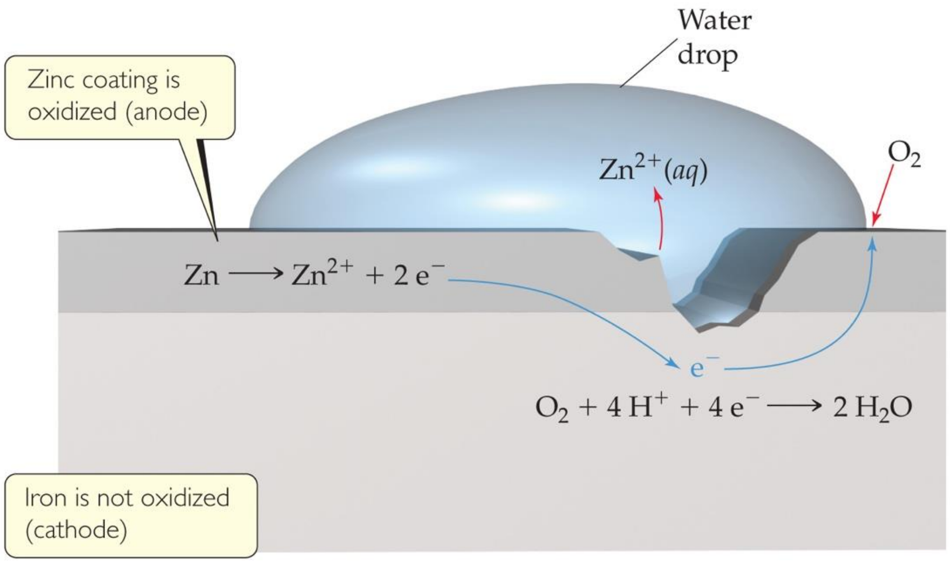


Corrosion



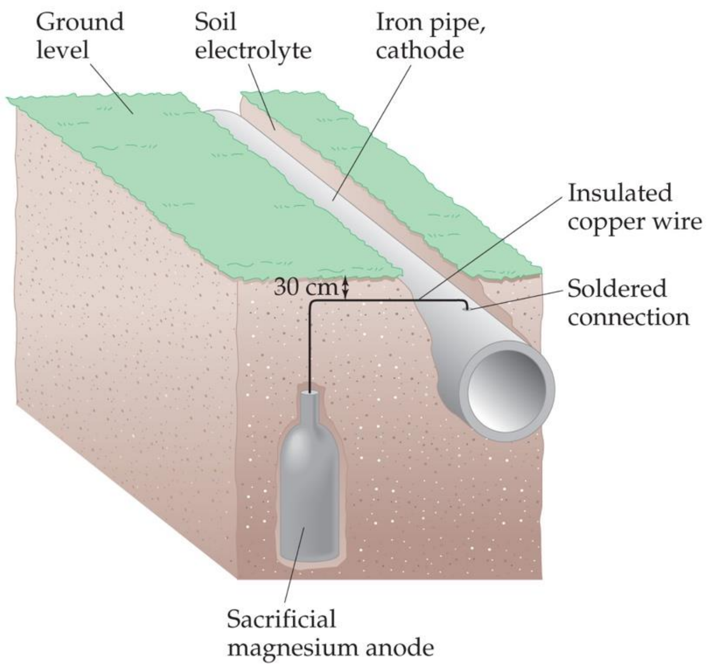
- Corrosion is oxidation.
- Its common name is *rusting*.

Preventing Corrosion



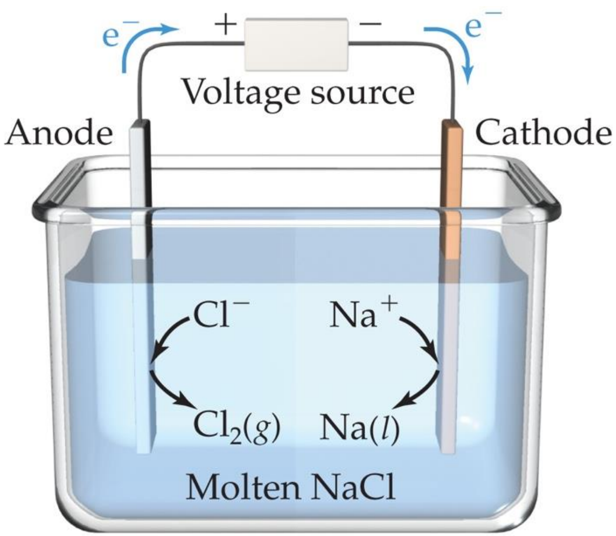
- Corrosion is prevented by coating iron with a metal that is more readily oxidized.
- **Cathodic protection** occurs when zinc is more easily oxidized, so that metal is sacrificed to keep the iron from rusting.

Preventing Corrosion



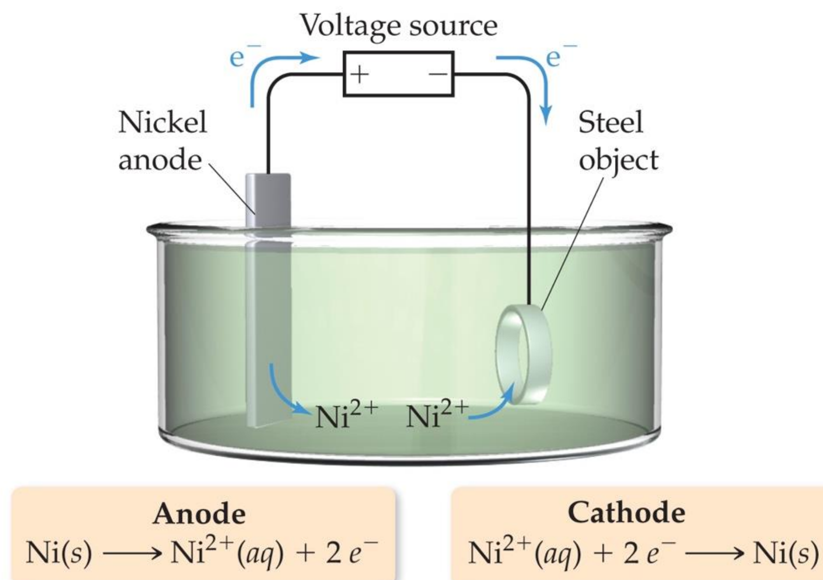
- Another method to prevent corrosion is used for underground pipes.
- A **sacrificial anode** is attached to the pipe. The anode is oxidized before the pipe.

Electrolysis



$2\,Cl^- \longrightarrow Cl_2(g) + 2\,e^-$

$2\,Na^+ + 2\,e^- \longrightarrow 2\,Na(l)$

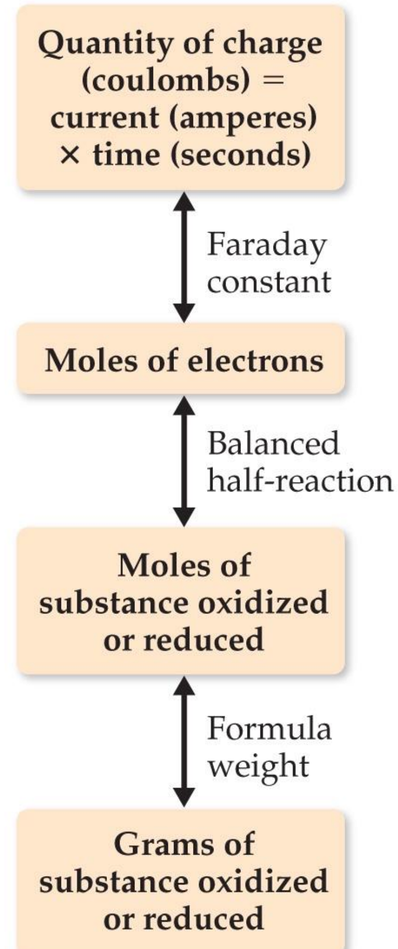


- Nonspontaneous reactions can occur in electrochemistry IF outside electricity is used to drive the reaction.
- Use of electrical energy to create chemical reactions is called **electrolysis**.

Electrolysis and "Stoichiometry"

- 1 coulomb = 1 ampere × 1 second
- $Q = It = nF$
- Q = charge (C)
- I = current (A)
- t = time (s)
- n = moles of electrons that travel through the wire in the given time
- F = Faraday's constant

NOTE: n is different than that for the Nernst equation!



Electrolysis and "Stoichiometry"

NOTE: *n* is different than that for the Nernst equation!

In the **Nernst equation**, **n** refers to the **number of electrons transferred in the balanced half-reaction** (e.g., for Cu^{2+} + $2e^{-} \rightarrow Cu$, **n = 2**).
In contrast, in **electrolysis**, **n** represents the **actual number of moles of electrons that have flowed** (e.g., if 10 moles of electrons pass through the circuit, **n = 10**).

→ Therefore, **n in the Nernst equation is a stoichiometric coefficient** (from the reaction equation),
while **n in electrolysis is a measurable quantity** (the actual moles of electrons transferred).